

M1: Monte Carlo and Nonlinear Spot Checks

4 October 2026

Do finite noisy records behave as the analytic engine predicts, away from p_0 ? Amir Dollie (DLLAMI004), EEE4022S SHARC buoy project Code: `runM1MonteCarloSpotChecks.m`, `m1Tools.m`, `g3EstimatorCore.m`, `resolveM1Scenarios.m` Results: `Results/M1/M1_20261003_221127` (commit 96e9b0a)

Summary

Question. D1, A1, R1 and R2 rest on *analytic* predictions (E1): the Fisher information F , the standard deviations $\sigma = \sqrt{\text{diag } F^{-1}}$ and the Welch-expected bias $\delta\theta_W$ of one record, without simulating records. G3 showed at p_0 that real finite records follow such predictions. Does that still hold in the four deliberately awkward cases fixed by the pre-registered rules? **Scope.** Truth and inverse share the EMM, so M1 verifies the estimator and the analytic Fisher/bias approximation, not the physical accuracy of the forward model (that is V1). **Outcome.** **All four cases pass every pre-registered criterion.** Empirical scatter over predicted lies between 0.94 and 1.09 (band 0.85 to 1.20), the 95% coverage of the predicted ellipse between 0.925 and 0.965 (band 0.88 to 0.99), and the nonlinear solver lands within 0.018σ of the linearised estimate (limit 0.2σ) from both $\beta \times 0.8$ and $\times 1.2$. The mean offsets are consistent with the Welch-expected bias (all within 1.4 standard errors). The weakest case is the Froude-scaled MIZ floe, whose scatter is 8 to 9% above prediction (1.4 to 1.7 standard errors), inside the band and without changing its A1 class.

1 What the study does

Why M1 exists. The analytic E1 engine replaced repeated Monte Carlo for D1, A1, R1 and R2. G3 validated the analytic route near p_0 , but those studies deliberately moved into harder parts of the design, forcing and floe domains. M1 tests whether the E1 predictions of Fisher uncertainty and Welch bias still describe real finite-record estimates in four pre-registered stress cases, each probing a different way the approximation could fail. It verifies the estimator and the analytic approximation, *not* the EMM physics: truth and inverse use the same forward model.

The four cases (fixed by rules written before the source studies ran; logged amendments after D1 and R2):

Case	Rule	Why it stresses the prediction	$\sigma_{\ln \beta}$	d_W	Class
D1	D1 optimum, $n_s = 2$ (s6, reference s18)	one transmissibility per bin	0.463%	0.216	Identifiable
A1	case nearest the Identifiable/Marginal boundary	furthest from p_0 ; Froude-scaled, 69 bins	1.005%	0.127	Marginal
R2	R2 stress sea ($\omega_p = 3.5$, $\gamma_J = 7$)	information from a weak spectral tail	0.766%	0.221	Identifiable
R1	p_0 at LSM6DSV16X noise	the measured-noise operating point	0.355%	0.309	Identifiable

Method (pre-registered in `cfg.M1`; clarification logged before the run).

- Stage 1.** Each case is rebuilt by its source study’s own code (`resolveM1Scenarios`) and evaluated by the frozen engine; it reproduced its source study exactly (relative difference $\leq 4 \times 10^{-16}$ in σ) before any record was generated.
- Stage 2.** 100 records per case from `synthesiseTwinRecords` (fixed seed schedule), corrected transmissibility features from G3’s estimator, and the linearised estimate $\Delta\theta = F^{-1}J^T\Sigma^{-1}(z - f_0)$ with the *analytic* J , Σ_k and F . Pass: empirical over predicted σ in $[0.85, 1.20]$ for both parameters.
- Stage 3.** $q = \Delta\theta^T F \Delta\theta$ (centred on the truth, as G3a, so $E[q] = 2 + d_W^2$) against χ_2^2 . Pass: 95% coverage in $[0.88, 0.99]$. The mean offset against $\delta\theta_W$ is a diagnostic, not a criterion. If a statistic’s 1-standard-error interval contains a limit, records 101 to 200 are added once (pre-registered “borderline” rule).
- Stage 4.** Record 1 inverted with G3’s safeguarded Gauss-Newton solver from $\beta \times 0.8$ and $\times 1.2$ (R true). Pass: converged and $|\theta_{NL} - \theta_{lin}|/\sigma < 0.2$ per parameter (G3.1).

Reuse, not reimplementation. The estimator, information matrix, objective, Jacobian and solver are G3’s own, copied by a program into `g3EstimatorCore.m`; the tester (12/12) checks every function body against the G3 source, and that the nonlinear model on the EMM reproduces the engine’s features at the A1 truth (2.6×10^{-15}). The records use the same node grid as the prediction, so node-grid mismatch is not a confound: M1 isolates the finite-record estimator and the analytic Fisher/bias prediction. The run refused to start with uncommitted code, so commit 96e9b0a is the code that ran (76 min).

2 Results

Case	n	emp/pred (± 1 SE)		coverage		mean q ($2 + d_W^2$)	largest NL – lin	Verdict
		$\ln \beta$	$\ln R$	68%	95%			
D1 $n_s = 2$	100	1.042 ± 0.074	1.080 ± 0.077	0.650	0.960	2.09 (2.05)	0.002σ	pass
A1 MIZ	200*	1.078 ± 0.054	1.091 ± 0.055	0.635	0.925	2.36 (2.02)	0.018σ	pass
R2 stress sea	200*	1.006 ± 0.050	0.957 ± 0.048	0.670	0.965	2.00 (2.05)	0.006σ	pass
R1 p_0 , LSM	100	0.939 ± 0.067	0.993 ± 0.071	0.680	0.940	2.03 (2.10)	0.004σ	pass

*Extended by the borderline rule: A1 on emp/pred, R2 on coverage at 100 records. No record lost an accepted bin. Nominal coverage 0.683 and 0.950.

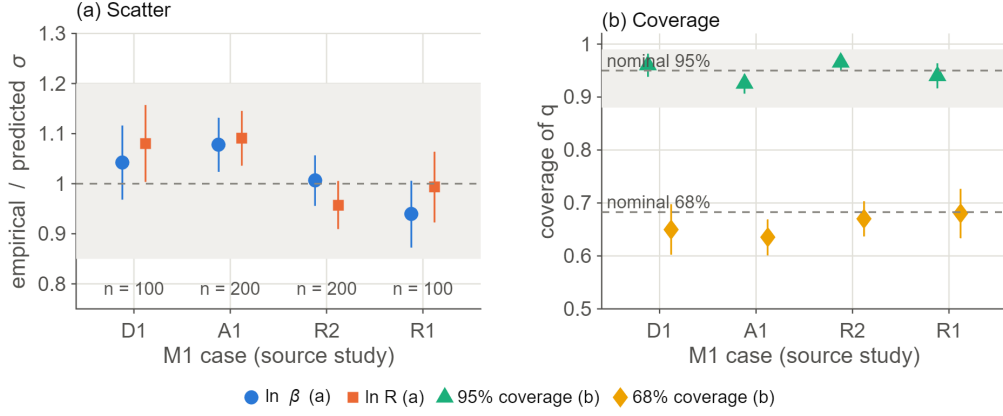


Figure 1. Acceptance per case. (a) Empirical over predicted standard deviation with ± 1 standard error; shaded: the pre-registered band. (b) Coverage of the predicted 68% and 95% regions with binomial ± 1 standard error; shaded: the 95% acceptance band.

2.1 Welch-bias prediction

Mean offset of the linearised estimates against the Welch-expected bias (a diagnostic, not a criterion; G3.3 rule: within 3 standard errors).

Case	$\ln \beta$ (%)			$\ln R$ (%)		
	Welch-expected	empirical mean $\pm$ SE	difference	Welch-expected	empirical mean $\pm$ SE	difference
D1	+0.097	+0.102 $\pm$ 0.048	+0.1 SE	+0.013	+0.025 $\pm$ 0.012	+0.9 SE
A1	+0.008	−0.013 $\pm$ 0.077	−0.3 SE	−0.021	−0.036 $\pm$ 0.014	−1.1 SE
R2	+0.143	+0.216 $\pm$ 0.055	+1.3 SE	+0.014	+0.026 $\pm$ 0.010	+1.2 SE
R1	+0.099	+0.133 $\pm$ 0.033	+1.0 SE	+0.011	+0.014 $\pm$ 0.007	+0.4 SE

No systematic bias discrepancy was resolved: all eight offsets are consistent with $\delta\theta_W$, the largest being R2 in $\ln \beta$ (0.216% against 0.143%, standard error 0.055%, 1.3 SE). This is compatible with the small analytic-versus-Monte-Carlo differences seen in the E1 validation, but M1 does not resolve such differences.

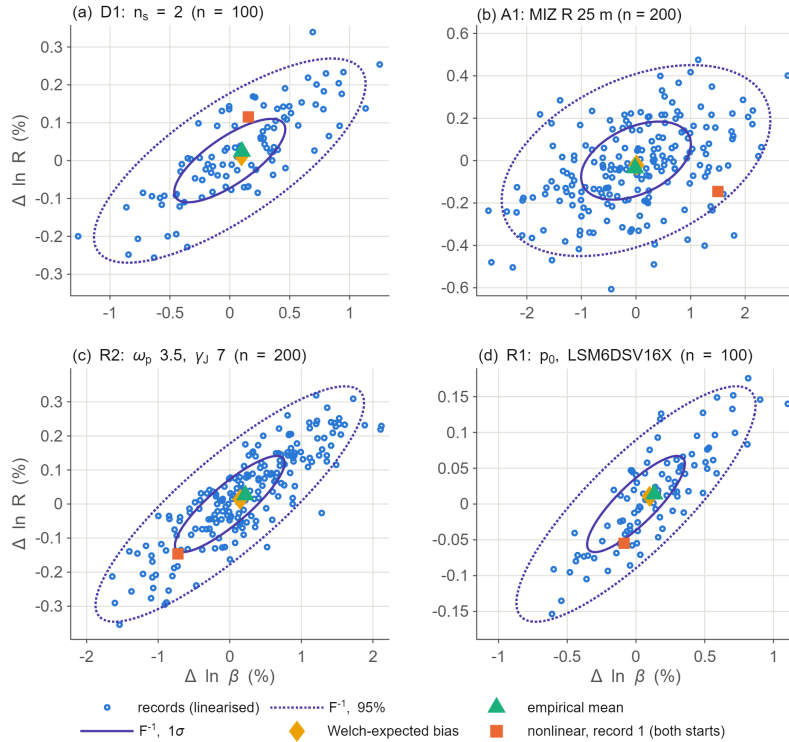


Figure 3. Per-record linearised estimates (minus the truth) against the analytic 1σ and 95% ellipses of F^{-1} , with the Welch-expected bias, the empirical mean and the nonlinear estimates of record 1 (both starts coincide). The strong β – R correlation predicted by F^{-1} is reproduced in every case.

2.2 Nonlinear spot checks

All eight runs converged in 3 to 5 iterations (13 to 26 model evaluations, 7 to 14 min each). The starts lie 18 to 63 predicted $\sigma_{\ln \beta}$ from the truth, yet both starts of each case land on the same estimate, within 1.5σ of the truth and at most 0.018σ from the linearised estimate of the same record; $\chi^2/\text{dof} = 0.88$ to 1.06. The largest gap is in the MIZ case, the least linear of the four.

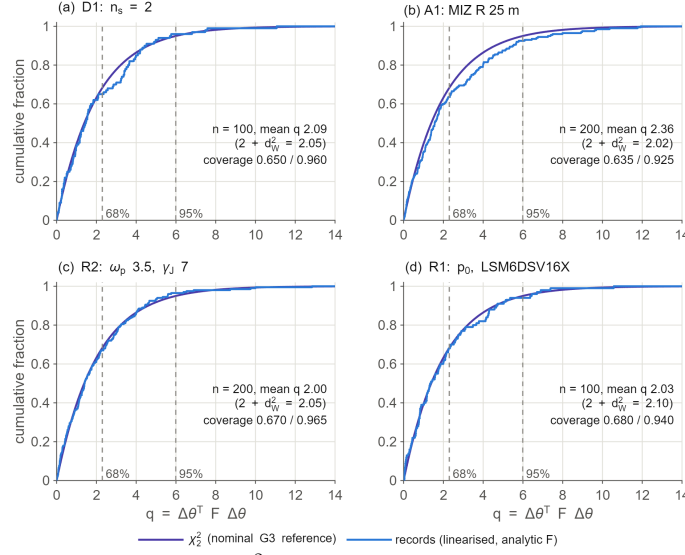


Figure 2. Empirical distribution of q per case against χ^2_2 , the nominal coverage reference of G3 (dashed: its 68% and 95% quantiles). Because q is centred on the truth and the estimator carries a small Welch bias, q is not exactly central χ^2_2 : its expected mean is $2 + d_W^2$ (given per panel).

3 Interpretation

1. **The analytic route holds where it was most likely to fail.** The minimal two-sensor design, the boundary floe far from p_0 , the weak-tail sea and the measured-noise operating point all reproduce the predicted scatter, ellipse and bias within sampling error. The analytic predictions behind D1, A1, R1 and R2 are therefore backed by finite-record simulation in their hardest cases, not only at p_0 (G3).
2. **No evidence of a systematic error in the predicted uncertainty.** The eight scatter ratios fall on both sides of 1 (0.94 to 1.09) with no consistent sign, and Figure 3 shows that the predicted correlation structure, not only the diagonal, is reproduced.
3. **The MIZ boundary case is the weakest.** Its empirical scatter is about 8 to 9% above prediction ($r_\beta = 1.078$, $r_R = 1.091$), and its elevated mean q (2.36 against 2.02) is consistent with that covariance inflation. It remains inside the pre-registered band and only about 1.5 standard errors from unity, so it is evidence of a possible mild optimism of the analytic σ rather than a demonstrated systematic error. It is the case furthest from p_0 (Froude-scaled, a narrower band of 69 bins). Even if the inflation were real, $\sigma_{\ln \beta}$ would move from 1.005% to about 1.08%, leaving the floe Marginal: A1's classification does not change.
4. **The Welch-bias prediction holds.** Every offset agrees with $\delta\theta_W$ within 1.4 standard errors; no systematic bias discrepancy is resolved.
5. **The inverse is well behaved within $\pm 20\%$ in β .** Starting tens of σ away, the solver converges quickly to the same point from both sides in every case, and that point is the linearised estimate. No second minimum was met, and the linear theory used everywhere else applies at the solution. Within the tested neighbourhood, failure of the inverse is therefore governed by information content and estimator bias, not by nonlinear optimisation pathology.

Conclusion. M1 verified the analytic inverse outside its original p_0 validation regime. Across the minimum two-sensor design, the A1 MIZ boundary, the R2 stress spectrum and the measured LSM6DSV16X noise, empirical scatter stayed within 9% of the Fisher prediction and the 95% coverage within the pre-registered interval. The largest deviation, for the MIZ boundary floe, suggests the analytic uncertainty may be about 8% optimistic there, without altering the A1 classification. All nonlinear inversions converged from $\pm 20\%$ stiffness starts and agreed with the linearised estimator within 0.02σ . This supports using the analytic E1 engine for the design, robustness and domain studies, while leaving forward-model validity to V1.

Limitations. Within-model: truth and inverse share the EMM, so M1 verifies the estimator and the Fisher/bias prediction, not the physics (A1 found the truncation unconverged at feature level; V1). Four deliberately selected stress cases: M1 spot-checks the difficult corners of the analytic programme; it is not an exhaustive Monte Carlo replication of every D1, A1, R1 and R2 scenario. 100 to 200 records resolve scatter ratios to about ± 5 to 7%, so discrepancies under about 10% are not resolved. The nonlinear check is one record per case from two starts in β only. Records use the prediction's node grid by design.

Requirement for the experiment and the thesis

Use the analytic Fisher and Welch-bias predictions for the design, robustness and domain conclusions; quote σ with a caveat of up to about 10% optimism for floes far from the basin reference (the MIZ cases). The linearised and nonlinear estimates are interchangeable at the solution, and the inverse converges from $\pm 20\%$ in β . The Conclusion above is the thesis wording.

Outputs (Results/M1/): M1_20261003.221127.log/.mat/_summary.txt; M1_analytic.csv, M1_summary.csv, M1_records.csv, M1_nonlinear.csv; checkpoints/; figures/M1_fig1 to M1_fig3 (.pdf/.png). Tester `testM1MonteCarloSpotChecks` 12/12.